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Sachin Vs World

Introduction

Sachin Tendulkar, widely hailed as the "God of Cricket," is a name synonymous with excellence and records in the realm of the sport. Among his numerous achievements, one of the most notable is his status as the leading run-scorer in Test cricket. However, this analysis delves deeper into whether Tendulkar's record is solely a product of the longevity of his career or if it truly signifies his supremacy over other cricketing legends.

Exploring Longevity vs. Performance:

Undoubtedly, Tendulkar's career spanned over two decades, affording him ample opportunities to accumulate runs. Yet, mere longevity cannot undermine the significance of his achievements. It is imperative to dissect Tendulkar's performance throughout his career to ascertain the validity of his standing as the best in Test cricket.

Statistical Analysis:

Delving into the statistics, Tendulkar's record of most international runs in the 5-day format is indeed remarkable. With 15,921 runs in 200 matches, his consistency and dominance are evident. However, it is essential to juxtapose his performance with other cricketing greats to gauge the true extent of his prowess.

Comparative Evaluation:

While Tendulkar's records are monumental, contemporaries like Ricky Ponting, Jacques Kallis, and Rahul Dravid also boast formidable statistics in Test cricket. Their contributions, albeit falling short of Tendulkar's aggregate runs, cannot be disregarded. Moreover, the impact of Tendulkar's innings on match outcomes and his ability to perform under pressure are factors that further elevate his stature.

Legacy Beyond Numbers:

Tendulkar's influence transcends mere statistics. His unparalleled skill, technique, and the manner in which he carried the hopes of a cricket-crazy nation for over two decades are unparalleled. Tendulkar's ability to adapt to various conditions and oppositions, coupled with his longevity, cements his legacy as one of the greatest cricketers of all time.

Dataset

Innings Player: The player who batted during the innings.

Innings Runs Scored: The total runs scored by the player in that innings.

Innings Runs Scored Num: Numerical representation of the runs scored by the player.

Innings Minutes Batted: The duration of time the player spent batting in that innings.

Innings Batted Flag: A flag indicating whether the player batted in that innings.

Innings Not Out Flag: A flag indicating whether the player remained not out at the end of the innings.

Innings Balls Faced: The number of balls faced by the batsman in that innings.

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Innings Boundary Fours: The number of boundary fours hit by the batsman in that innings.

Innings Boundary Sixes: The number of boundary sixes hit by the batsman in that innings.

Innings Batting Strike Rate: The batting strike rate of the player in that innings (calculated as runs scored divided by balls faced, multiplied by 100).

Innings Number: The sequential number of the innings within a match.

Opposition: The opposing team against which the innings was played.

Ground: The venue where the innings took place.

Innings Date: The date on which the innings was played.

Country: The country to which the player belongs.

50's: The number of half-centuries (50 runs) scored by the player.

100's: The number of centuries (100 runs) scored by the player.

Innings Runs Scored Buckets: Categorization of the runs scored by the player in ranges (buckets).

Innings Overs Bowled: The number of overs bowled by the player.

Innings Bowled Flag: A flag indicating whether the player bowled in that innings.

Innings Maidens Bowled: The number of maidens (overs with no runs scored) bowled by the player.

Innings Runs Conceded: The total runs conceded by the bowler in that innings.

Innings Wickets Taken: The total number of wickets taken by the bowler in that innings.

4 Wickets: The number of times the bowler took 4 wickets in an innings.

5 Wickets: The number of times the bowler took 5 wickets in an innings.

10 Wickets: The number of times the bowler took 10 wickets in a match.

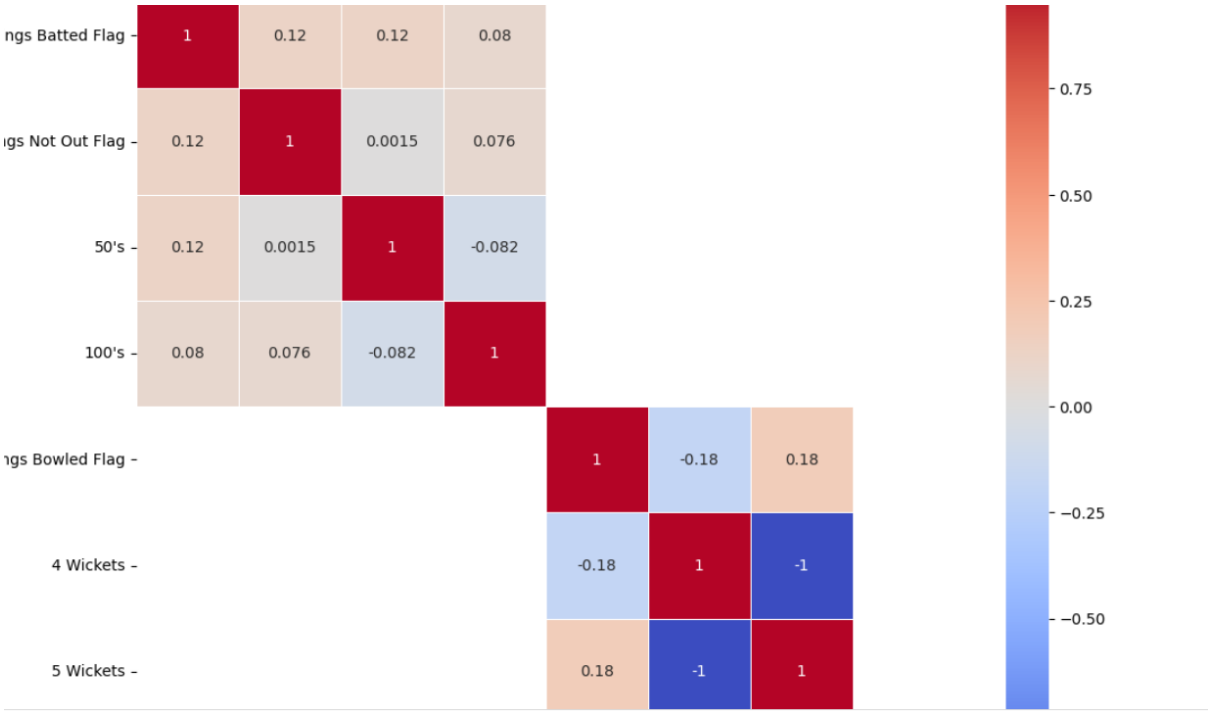
Innings Wickets Taken Buckets: Categorization of the wickets taken by the bowler in ranges (buckets).

Innings Economy Rate: The economy rate of the bowler in that innings (calculated as runs conceded divided by overs bowled).

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Pre-processing Dataset

1) By plotting the confusion matrix we could see that bowling related data was high uncorrelated to rest of the dataset.



Therefore, we dropped those columns.

```
  # Dropping values
df.drop(['Innings Overs Bowled','Innings Bowled Flag','Innings Maidens Bowled','Innings Runs Conceded','Innings Wickets Taken','4 Wickets','5 Wickets','10 Wickets','Innings Wickets Taken Buckets','Innings Economy Rate'], axis=1,inplace=True)
```

2) Trimming: Remove outliers based on statistical measures

```
import pandas as pd
import numpy as np

# Assuming newdf is your DataFrame with the column "Innings Batting Strike Rate"

# Convert the column to numeric type if it's not already numeric
newdf['Innings Batting Strike Rate'] = pd.to_numeric(newdf['Innings Batting Strike Rate'], errors='coerce')

# Calculate z-scores for the column
z_scores = np.abs((newdf['Innings Batting Strike Rate'] - newdf['Innings Batting Strike Rate'].mean()) / newdf['Innings Batting Strike Rate'].std())
```

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```
# Threshold for z-score based outlier removal (e.g., threshold = 3 for
3 standard deviations)
z_score_threshold = 3

# Mask to identify outliers based on z-scores
outlier_mask = z_scores > z_score_threshold

# Alternatively, you can use the interquartile range (IQR) method for
outlier removal
# Calculate the first quartile (Q1) and third quartile (Q3)
Q1 = newdf['Innings Batting Strike Rate'].quantile(0.25)
Q3 = newdf['Innings Batting Strike Rate'].quantile(0.75)
# Calculate the IQR
IQR = Q3 - Q1
# Threshold for IQR-based outlier removal (e.g., threshold = 1.5)
IQR_threshold = 1.5
# Mask to identify outliers based on IQR
outlier_mask_iqr = (newdf['Innings Batting Strike Rate'] < (Q1 -
IQR_threshold * IQR)) | (newdf['Innings Batting Strike Rate'] > (Q3 +
IQR_threshold * IQR))

# Remove outliers from the DataFrame using the mask
# For z-score method
newdf_no_outliers_z_score = newdf[~outlier_mask]
# For IQR method
newdf_no_outliers_iqr = newdf[~outlier_mask_iqr]

# Print the DataFrame without outliers for both methods
print("DataFrame without outliers (Z-score method):")
print(newdf_no_outliers_z_score)
print("\nDataFrame without outliers (IQR method):")
print(newdf_no_outliers_iqr)
```

Here we showcases two distinct methods for detecting and removing outliers from a DataFrame containing cricket data, particularly focusing on the "Innings Batting Strike Rate" column. The first method employs z-scores, which standardizes the values and identifies outliers based on a predefined threshold, typically set at 3 standard deviations from the mean. Alternatively, the IQR method utilizes quartiles to define a range within which most data points lie, flagging values outside this range as outliers. After detecting outliers using both methods, the code filters the DataFrame to exclude these anomalous data points. This systematic approach ensures the integrity of the dataset, facilitating more accurate analyses and insights into cricket performance metrics.

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3) Feature Engineering

```
newdf['Innings Year'] = pd.DatetimeIndex(newdf['Innings Date']).year
newdf['Innings Boundary'] = newdf['Innings Boundary Fours'] +
newdf['Innings Boundary Sixes']
```

```
# prompt: Using dataframe newdf: convert Innings Boundary Fours and
Innings Boundary Sixes from str to numeric

import pandas as pd
newdf['Innings Boundary Fours'] = pd.to_numeric(newdf['Innings Boundary
Fours'])
newdf['Innings Boundary Sixes'] = pd.to_numeric(newdf['Innings Boundary
Sixes'])
newdf['Innings Runs Scored Num'] = pd.to_numeric(newdf['Innings Runs
Scored Num'])
```

```
# prompt: Using dataframe newdf: multiple Innings Boundary Fours by 4
and Innings Boundary Sixes by 6 and divide it by Innings Boundary Sixes
into new column called boundary percentage

newdf['boundary_percentage'] = (newdf['Innings Boundary Fours'] * 4 +
newdf['Innings Boundary Sixes'] * 6) / newdf['Innings Runs Scored Num']
```

ındary_percentage
0.44898

The DataFrame "newdf" is manipulated to derive additional meaningful columns related to cricket statistics. Initially, the 'Innings Year' column is created by extracting the year component from the 'Innings Date' column using the Pandas DateTimeIndex functionality, facilitating analysis based on yearly trends. Subsequently, a new column 'Innings Boundary' is formed by summing the counts of fours and sixes hit in an innings, consolidating boundary-hitting data. Following this, string-type columns 'Innings Boundary Fours', 'Innings Boundary Sixes', and 'Innings Runs Scored Num' are converted to numeric types for numerical operations. Finally, a 'boundary_percentage' column is generated by calculating the percentage of runs scored through boundaries, computed by multiplying fours by 4,

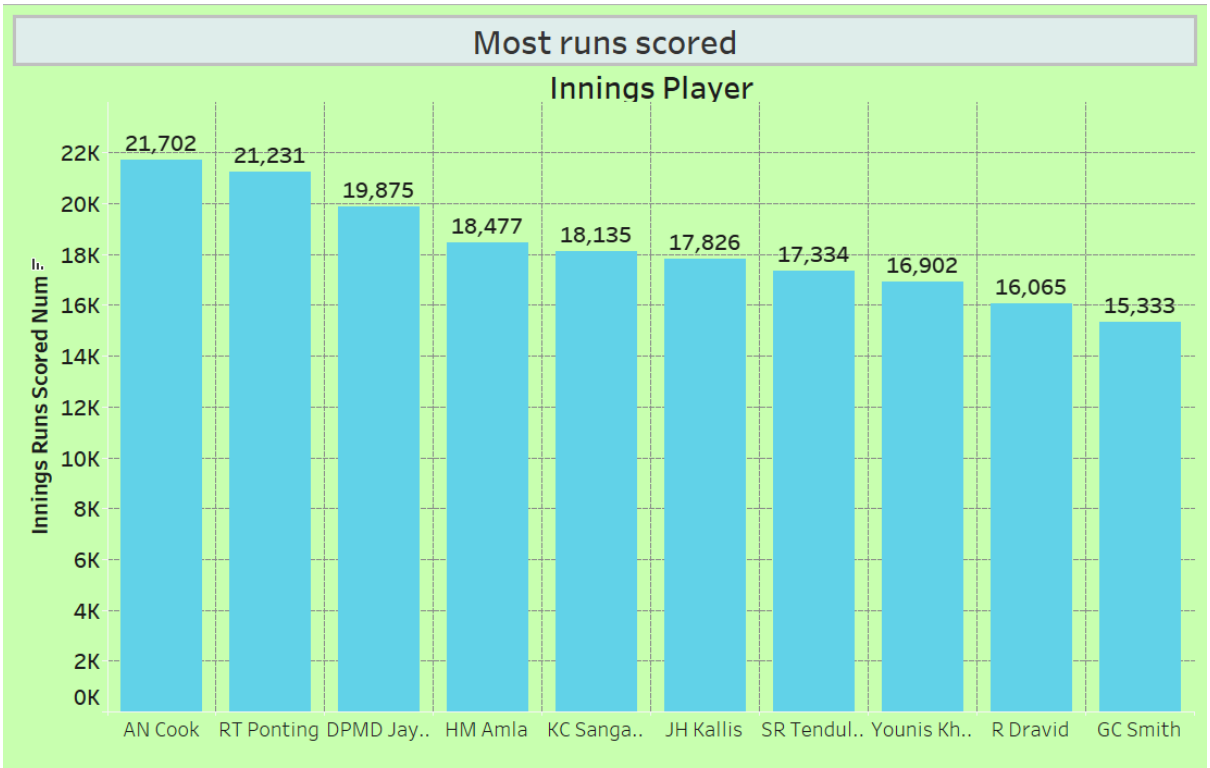
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adding it to sixes multiplied by 6, and dividing by the total runs scored, enabling deeper insights into the scoring patterns of cricket innings. This series of transformations enhances the analytical capabilities of the DataFrame, facilitating comprehensive cricket data analysis.

Analysis and Visualization

1) Runs Scored
   Numerical representation of the runs scored by the player.

   Categories
   a) We found out runs scored for all players. For more clarity we applied the filter for opposition as against Australia, England, India, New Zeeland, and South Africa.

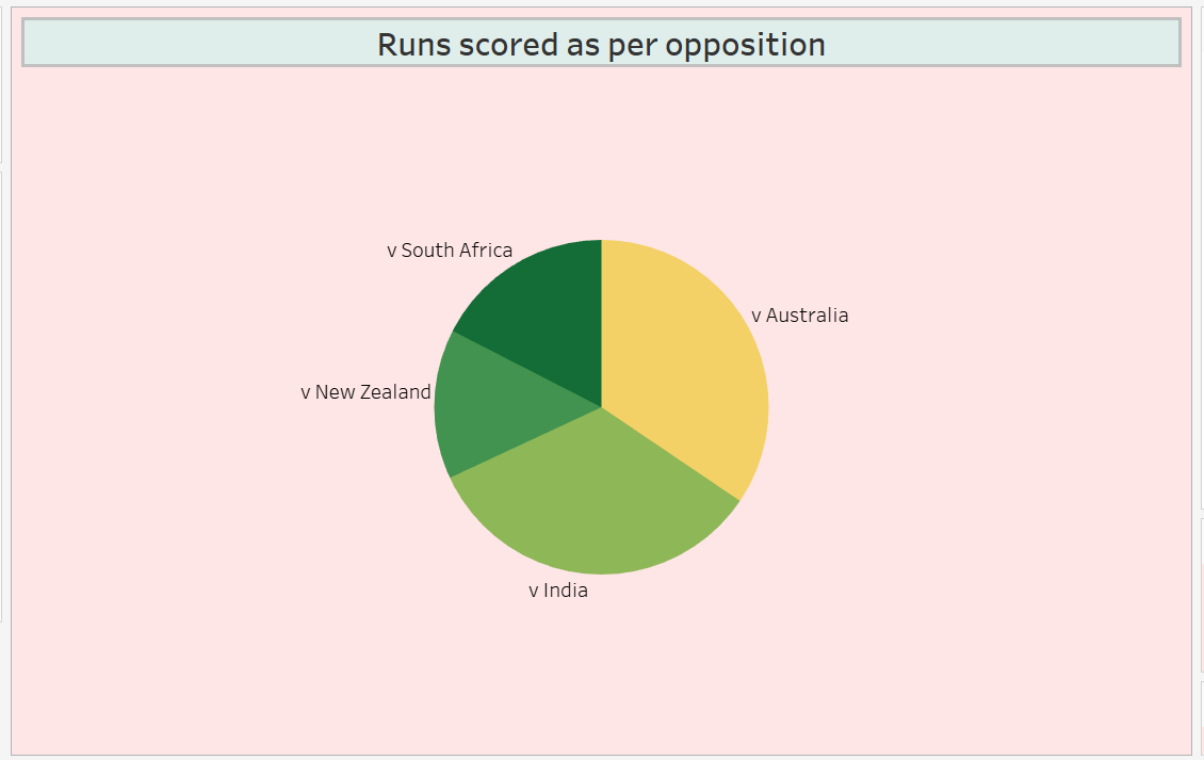


From this visual we found out Alistar Cook has scored most runs in the red ball format.

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b) Opposition

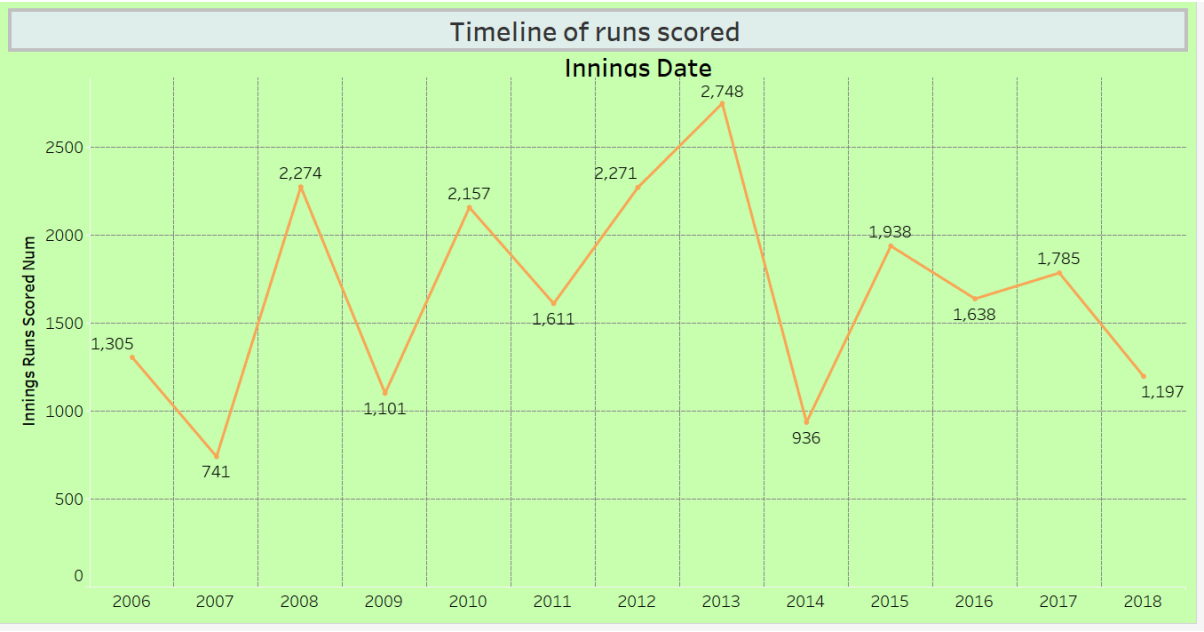
Then we found out runs distribution for Cook as per the opposition



For the pie chart we can conclude the Cook has scored most of his runs against India and Australia.

c) Year

Then we found progression of Cook's red ball career by year

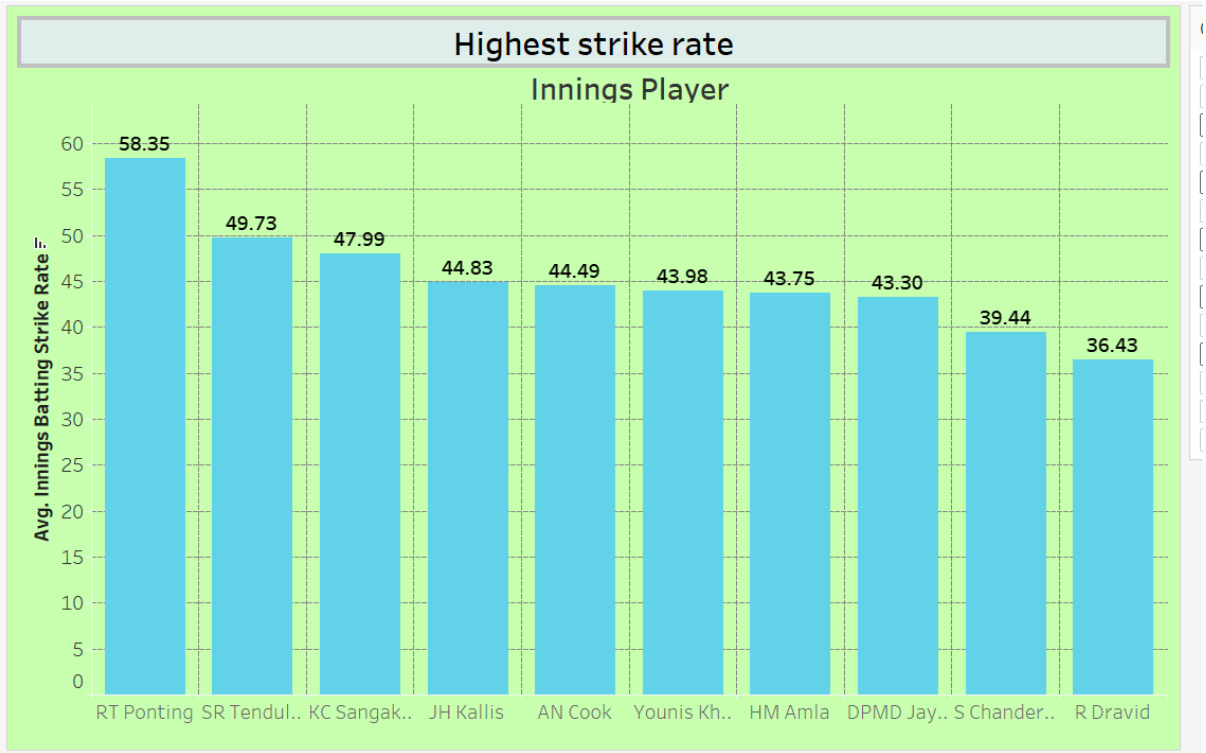


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Cook's best year – 2013
Cook's Worst year - 2007

2) Strike Rate
The batting strike rate of the player in that innings (calculated as runs scored divided by balls faced, multiplied by 100).

Categories

a) We found out highest innings strike rate for all players. For more clarity we applied the filter for opposition as against Australia, England, India, New Zeeland and South Africa.
For avoided anomalies as per player who have had small career we applied a filter to visualize for only the player who have batted the most in red ball format.

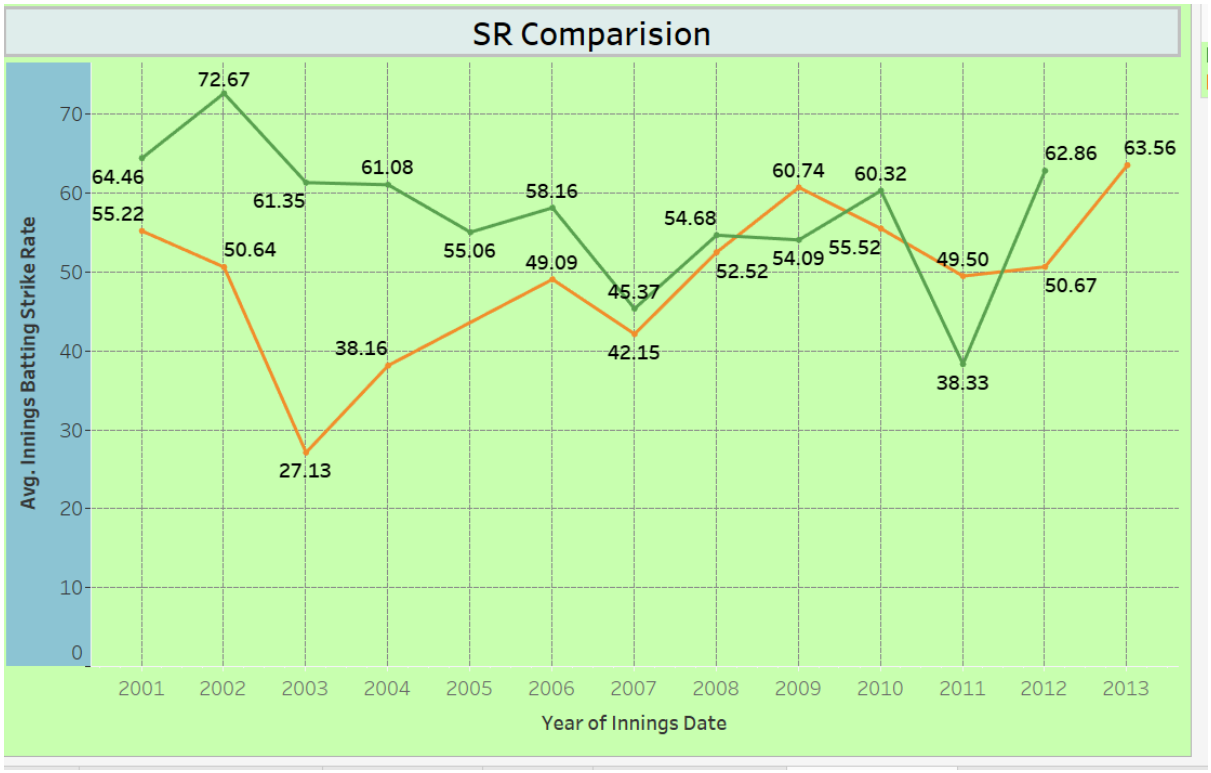


For the visual we can conclude the Ponting and Tendulkar are the two most fastest player in red ball cricket.

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b) Year

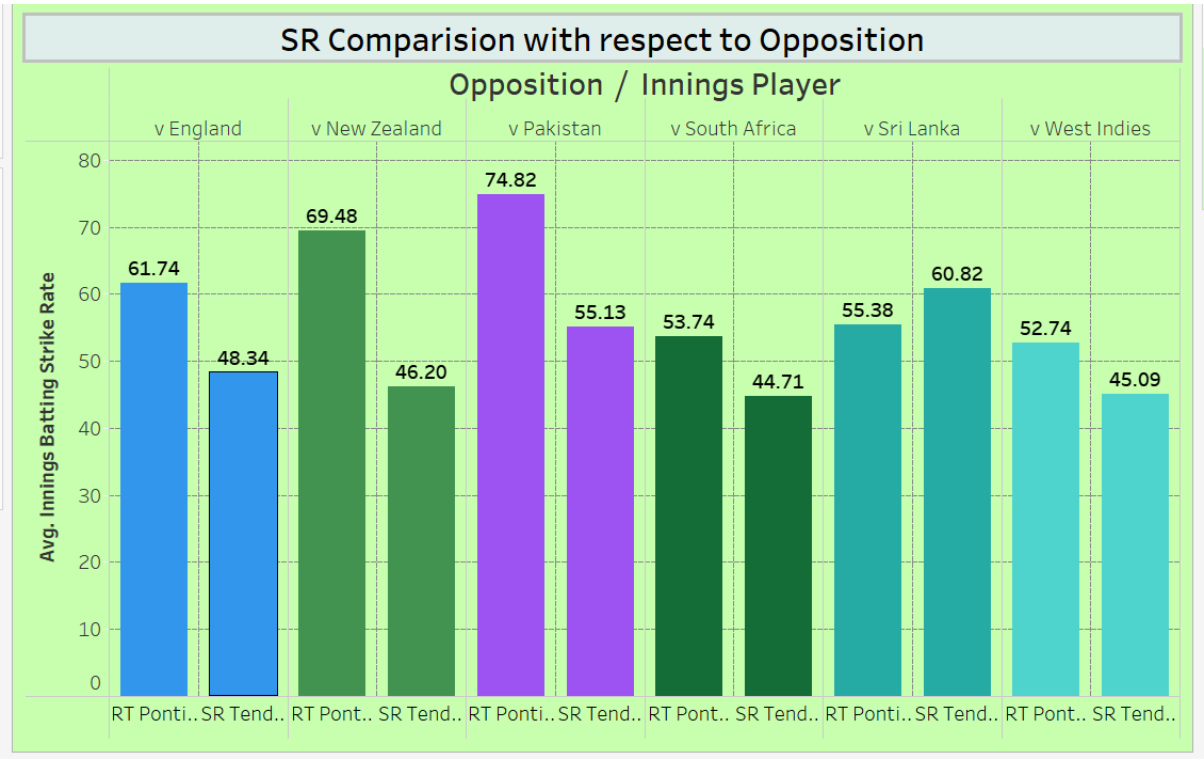
Then we compared both Sachin and Ponting strike rate with the progression of time.



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c) Opposition

We then compared both with respect to opposition



Baring Sri Lanka Ponting has higher strike Rate against all opposition. It can be traced back Ponting's struggle against Muralitharan and Sachin excellence for playing quality spin.

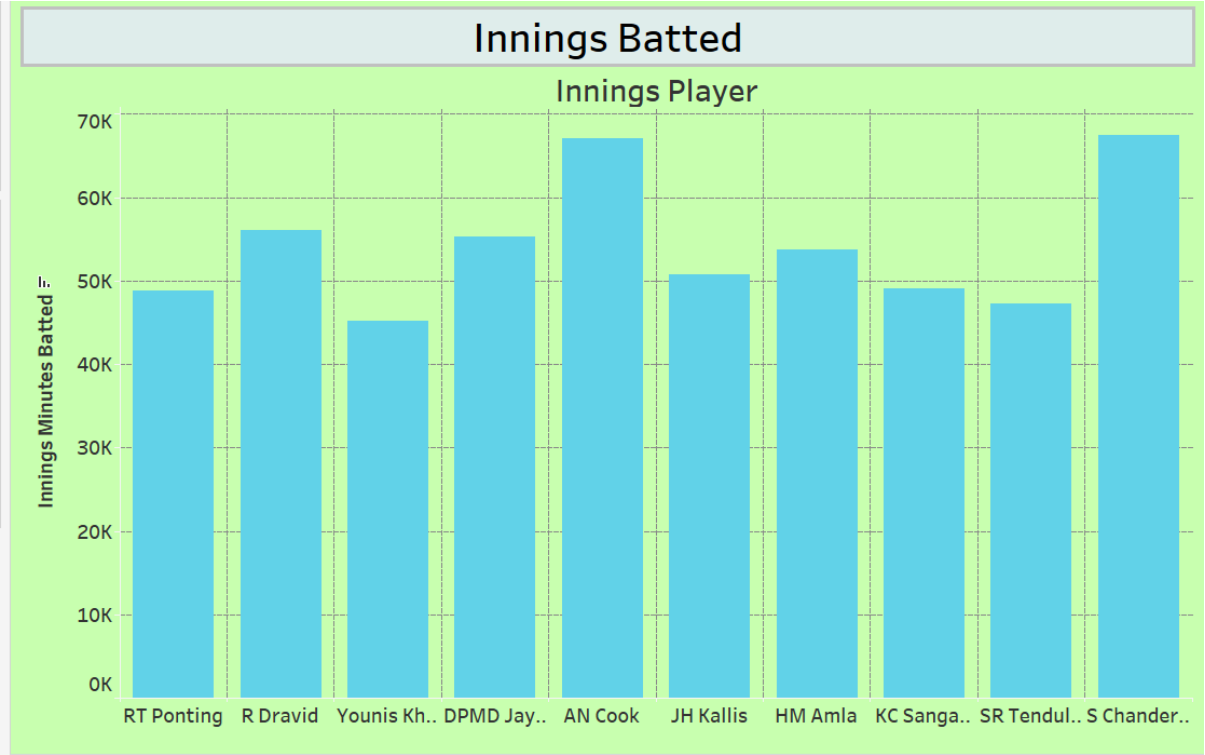
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3) Minute Batted

Red ball format is not only the measure for who can make most runs. It also test ones temperament to bat for long periods

Categories

a) Player
We found out minute batted for all players. For more clarity we applied the filter for opposition as against Australia, England, India, New Zeeland, and South Africa.

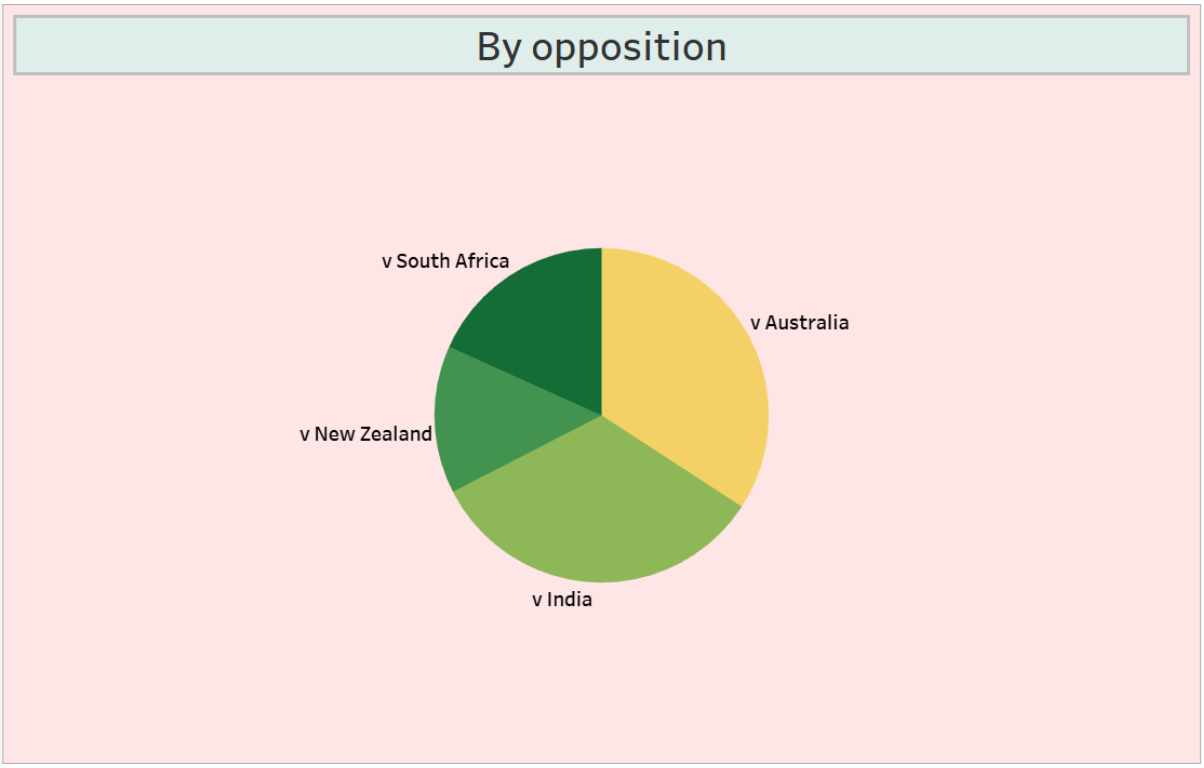


Alistar Cook exceed this visual as well.

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b) Opposition

Then we distributed it as per opposition

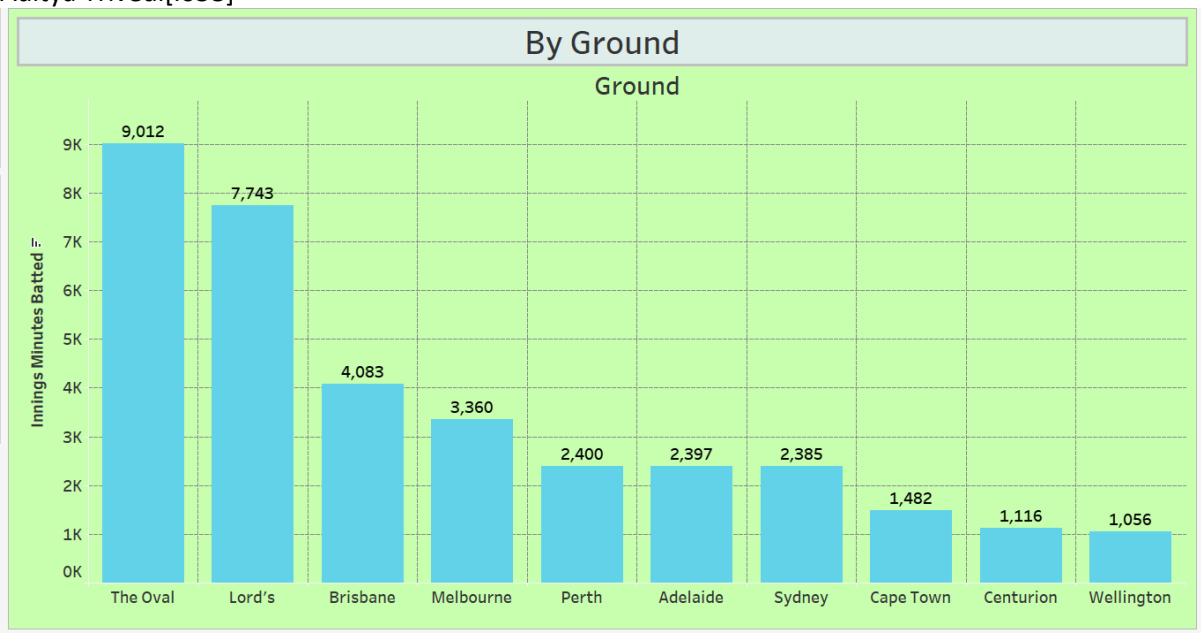


Similarly, as runs Cook has batted most against India and Australia.

c) Ground

Then we found out distribution of minute batted as per venues.

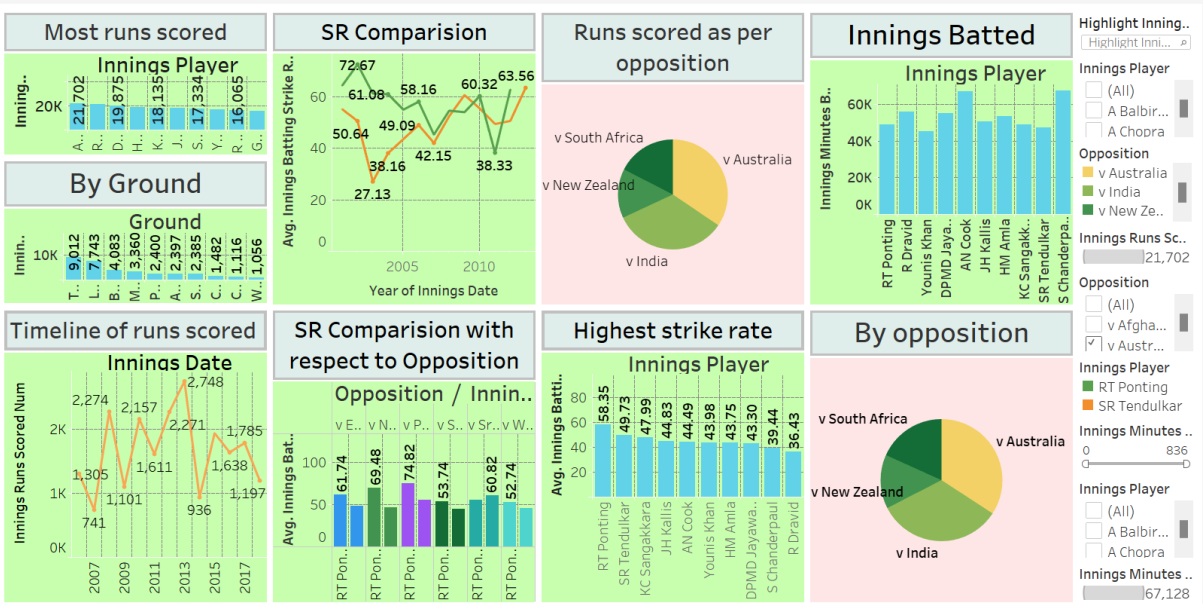
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We can see all the venues supporting pace balling high but none ailing spin in top 10 which suggests Cook's struggle against spin.

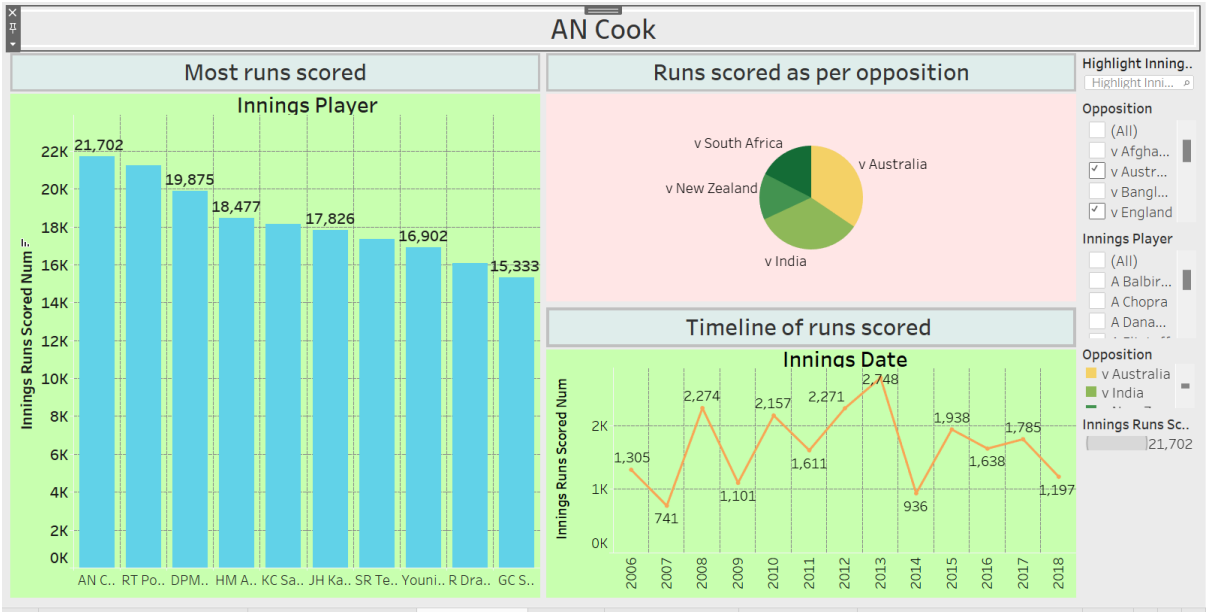
Dashboard and Story

Overall Dashboard

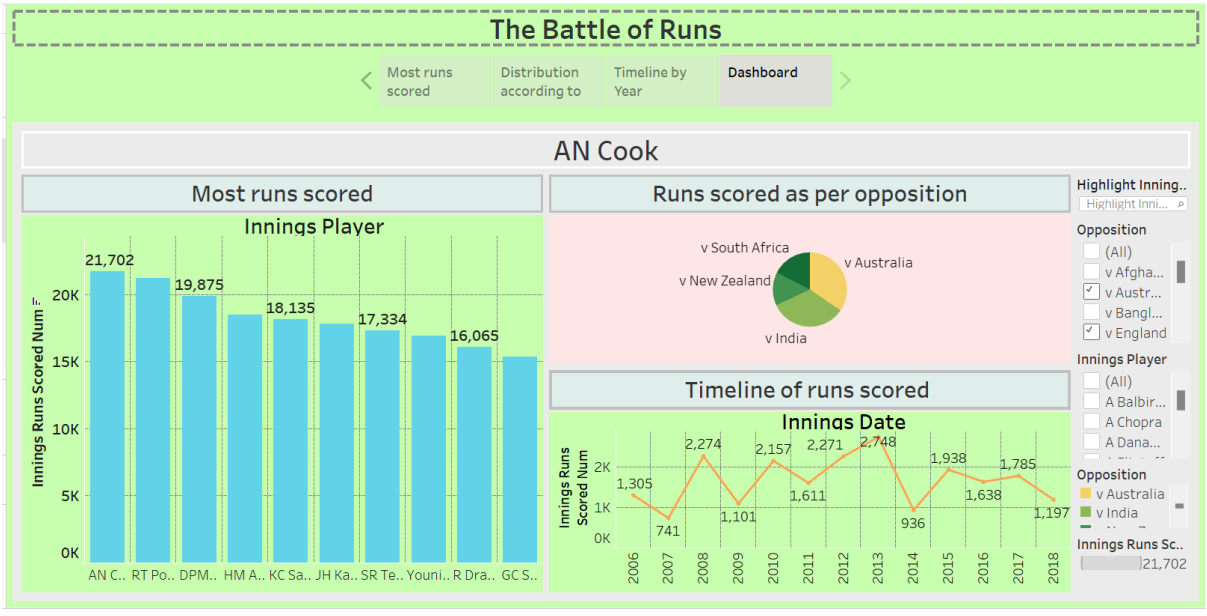


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Dashboard 1

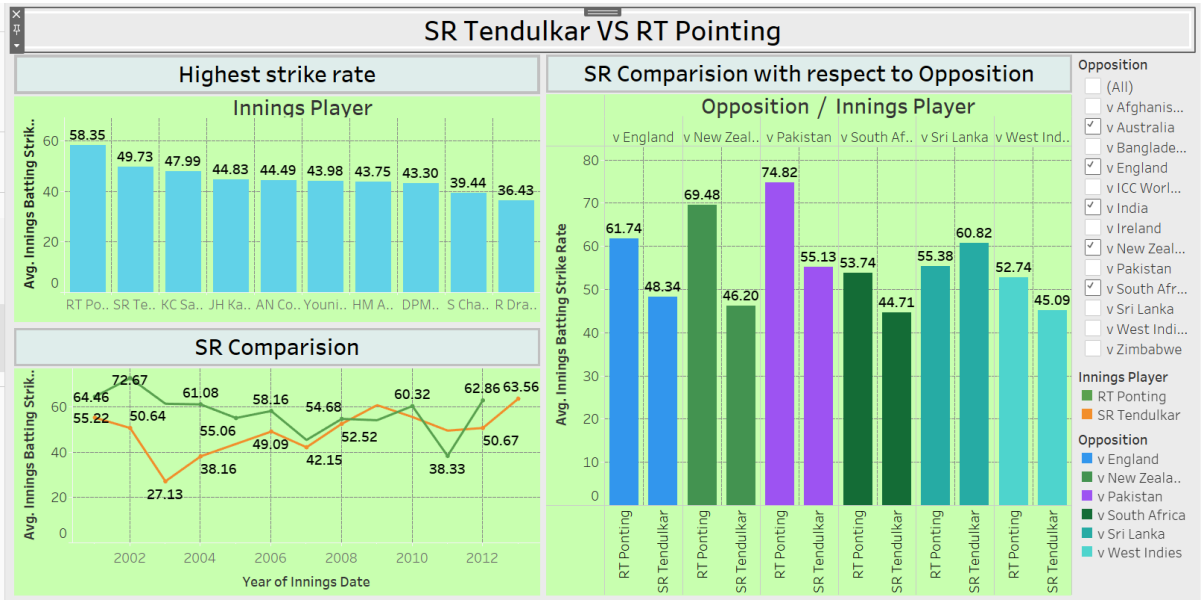


Story 1

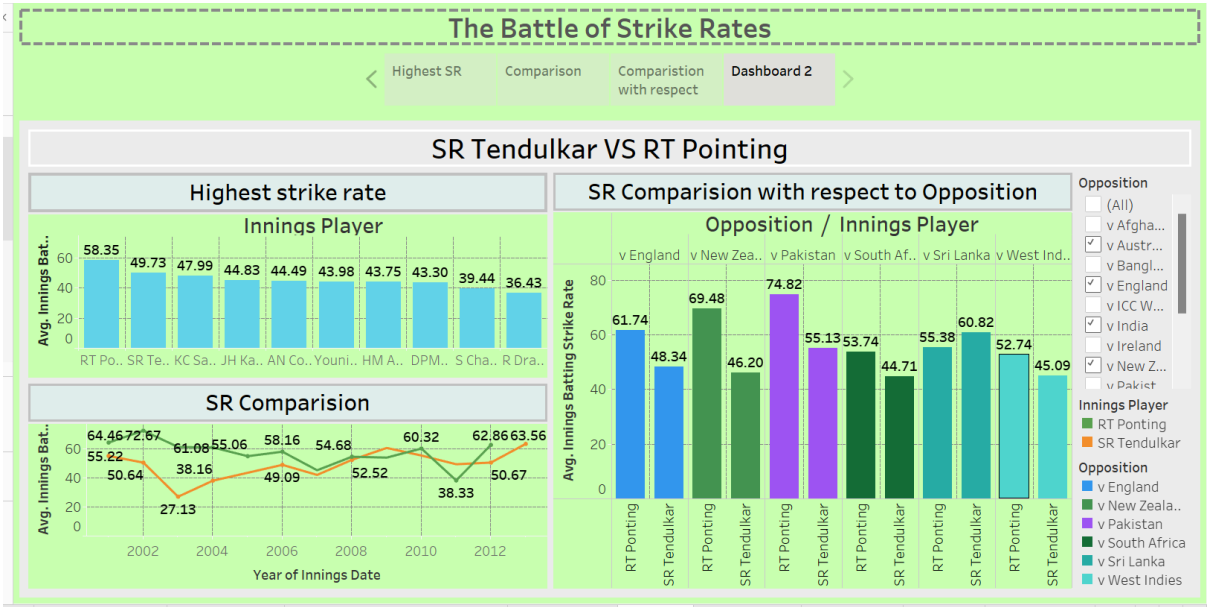


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Dashboard 2

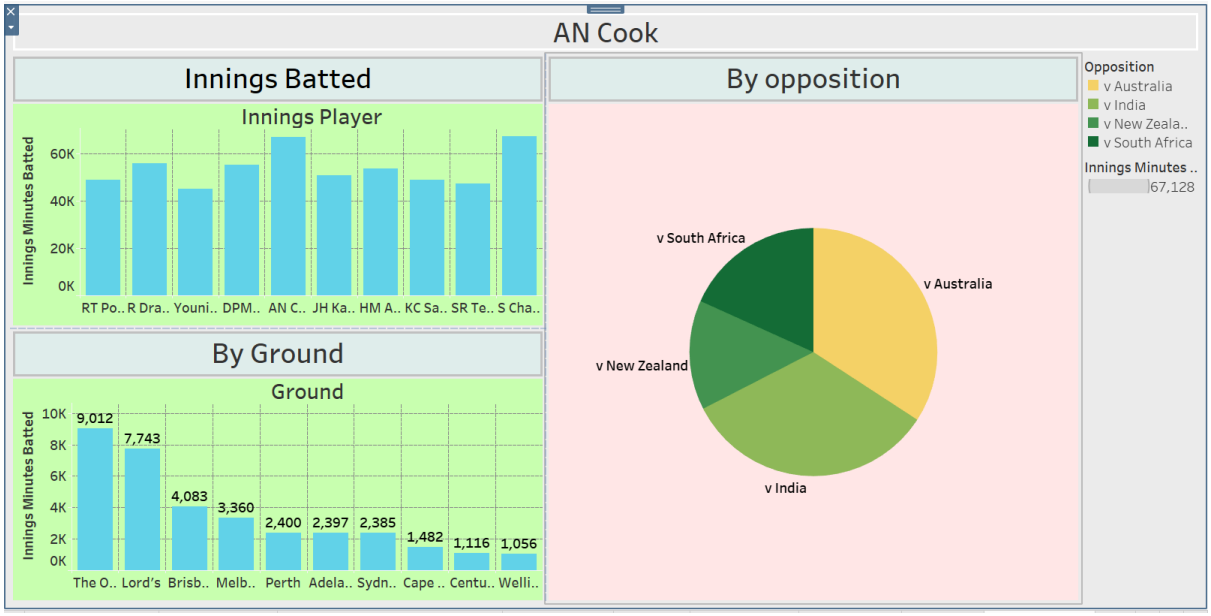


Story 2

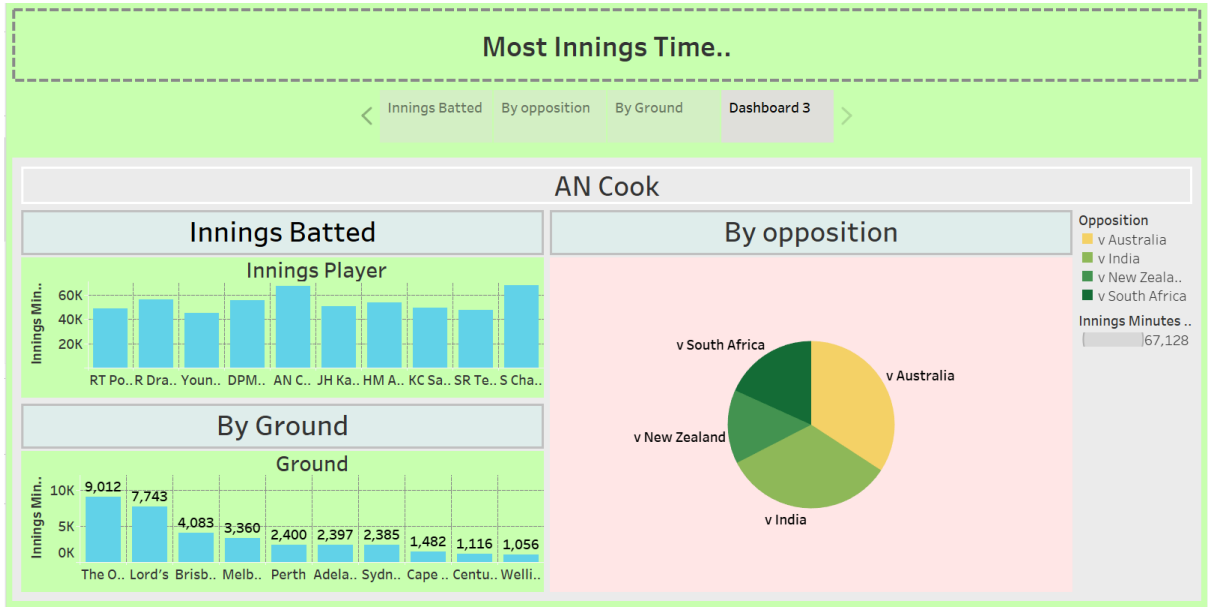


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Dashboard 3



Story 3



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Conclusion

From the Analysis we can conclude that 200 test and additional first-class matches played is a big factor which puts Sachin above all the players in his era.
Cook has more runs in red ball format, but he played lots of first class cricket after retirement.
He has batted longer, but it was his role as an opener.
Ponting has high Strike Rate, but the difference is not so big.

At the end of the day number cannot define a player completely it is the aura the a player had during their career which was certainly high for Sachin as he the backbone of Indian Batting lineup.

In conducting a comprehensive analysis of cricketing legends, particularly focusing on Sachin Tendulkar and his peers, several key facets emerge that delineate the nuances of their respective legacies. Undoubtedly, Tendulkar's remarkable feat of participating in 200 test matches, alongside an extensive repertoire of first-class encounters, underscores the sheer magnitude of his career. This longevity sets him apart, symbolizing an era-defining dominance that few can emulate. While Alastair Cook may hold a greater volume of runs in the red-ball format, it's crucial to contextualize this achievement, considering his continued participation in first-class cricket post-retirement and the inherent advantages of his role as an opener. Similarly, Ricky Ponting's formidable strike rate, while commendable, doesn't markedly overshadow Tendulkar's, highlighting the subtle yet significant differences in their approaches to the game. However, beyond the realm of statistics lies the intangible essence of a player's aura, a quality Tendulkar exuded effortlessly. As the linchpin of the Indian batting lineup for decades, his influence transcended numerical metrics, leaving an indelible mark on the sport's fabric. Ultimately, while numbers offer a glimpse into a player's prowess, it's the enduring impact and reverence garnered throughout a career that truly define greatness. In this regard, Sachin Tendulkar's legacy remains unrivaled, serving as a beacon of inspiration for generations of cricketers and fans alike.

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